Math 3
10-2D Solve Equations Algebraically: Word Problems PRACTICE

Name:_____
Date: _____ Per: ___

1) Belda is renting a boat to sail across Lake Hylia. She can rent a boat for a boat fee of 200 rupees as a cost of 15 rupees per hour. She has 300 rupees.

Write and solve an equation to determine the number of hours she can rent the boat before she needs to return it.

[$ PER HOUR] · [HRS] + [FEE] = [TOTAL]

LET X = # HRS

$15x + 200 = 300$

$-200 \quad -200$

$\frac{15x}{15} = \frac{100}{15}$

$x = 6\frac{2}{3}$ HRS

2) Bingle is going to a theme park. Admission is $150 and churros cost $8 each. Souvenirs cost $25 each. Bingle has $600 and plans to purchase 2 souvenirs then spend the rest of his money on churros.

Write and solve an equation to determine how many churros he can purchase.

[ADMIT] + [CHURRO $] + [SOUV $] = [TOTAL]

LET X = # CHURROS PURCHASED

$150 + 8x + 2(25) = 600$

$150 + 8x + 50 = 600$

$8x + 200 = 600$

$-200 \quad -200$

$\frac{8x}{8} = \frac{400}{8}$

$x = 50$ CHURROS

For problems 3 - 5, use the information below

Blink wants to open a hylian bike repair shop. He needs 800 rupees to buy tools and equipment then will need to spend 1400 rupees a month to keep the business running. Blink has 420,800 rupees saved up.

3) Write and solve an equation to determine how many months Blink can keep the shop open without earning more money.

[TOOLS] + [$ MONTHLY][MONTHS] = [TOTAL]

LET X = # MONTHS

$800 + 1400x = 420800$

$-800 \quad -800$

$\frac{1400x}{1400} = \frac{420000}{1400}$

$x = 300$ MONTHS

4) Determine how much Blink will have spent after 3 months.

(TOOLS) + (MONTHLY)(# MONTHS)

$800 + 1400(3)$

$800 + 4200$

5000 RUPEES

5) After 3 months, Blink has earned ~~5,000~~ 6000 rupees. Use the equation P = R - C where 'P' represents profit, 'R' represents revenue, and 'C' represents costs to determine Blink's profit.

$P = 6000 - 5000$

$= 1000$ RUPEES

6) Banon is looking at a triangular object. He was told the longest side is 30 cm longer than ⅓ of the perimeter. The shortest side is ¼ of the perimeter. He is finally told that the longest side is 60 cm longer than the shortest side.

Write and solve an equation to determine the perimeter of this object.

[LONGEST] = [SHORTEST] + 60

LET P = PERIMETER

$12\left[\frac{1}{3}P + 30\right] = \left[\frac{1}{4}P + 60\right]12$

$4P + 360 = 3P + 720$

$-3P \quad -3P$

$P + 360 = 720$

$-360 \quad -360$

$P = 360$ CM

For problems 7 and 8, utilize the information below:

Bobra is looking to sign up for a streaming service. Service A costs $10 a month if you pay a $20 sign up fee. Service B costs $15 a month.

7) Write and solve an equation to determine the number of months it would take for both streaming services to cost the same amount.

[SERVICE A] = [SERVICE B]

LET X = # MONTHS

$10X + 20 = 15X$

$-10X \quad -10X$

$\frac{20}{5} = \frac{5X}{5} \rightarrow X = 4$ MONTHS

8) Which service would you recommend Bobra sign up for? Explain.

SERVICE A, ITS COST PER MONTH IS CHEAPER, SO IT WILL BE CHEAPER IN THE LONG RUN

For problems 9 and 10, use the information below:

In the 2013 NFC Divisional Round the 49ers scored 26 more points than ¼ the total points scored in the game. The Packers scored ~~3 more~~ 7 less than ½ the total points scored in the game.

9) Write and solve an equation to determine the total number of points in the game.

TOTAL = 49ER + PACKERS

LET X = TOTAL POINTS

$4(X) = \left[\left(\frac{1}{4}X + 26\right) + \left(\frac{1}{2}X - 7\right)\right]4$

$4X = X + 104 + 2X - 28$

$4X = 3X + 76$

$-3X \quad -3X$

$X = 76$ POINTS

10) Who lost this game? By how many points did they lose?

49ERS	PACKERS
$\frac{1}{4}X + 26$	$\frac{1}{2}X - 7$
$\frac{1}{4}(76) + 26$	$\frac{1}{2}(76) - 7$
$19 + 26$	$38 - 7$
45 PTS	31 PTS

PACKERS LOST BY 14

ACTUAL PIC FROM THE GAME! →

